Name : Om Sutar
Organization : CyArt
Date : 20th March, 2026.

Title: SOC Implementation and Threat Analysis using Wazuh

Introduction

Clear and well-structured documentation is essential for understanding Security Operations Center (SOC) workflows. Visuals such as dashboards, logs, and event details enhance comprehension of security monitoring and incident response. This document demonstrates practical SOC operations using Wazuh, including threat hunting, alert triage, incident response, and analysis. The intended audience includes cybersecurity students and professionals seeking hands-on SOC experience.

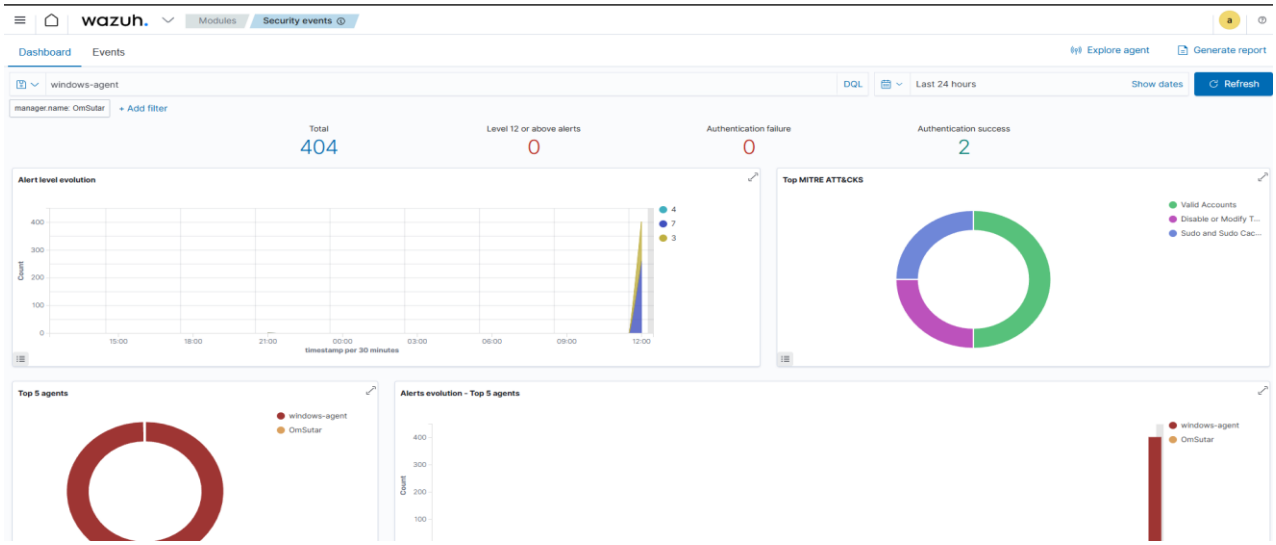
1. Threat Hunting Practice

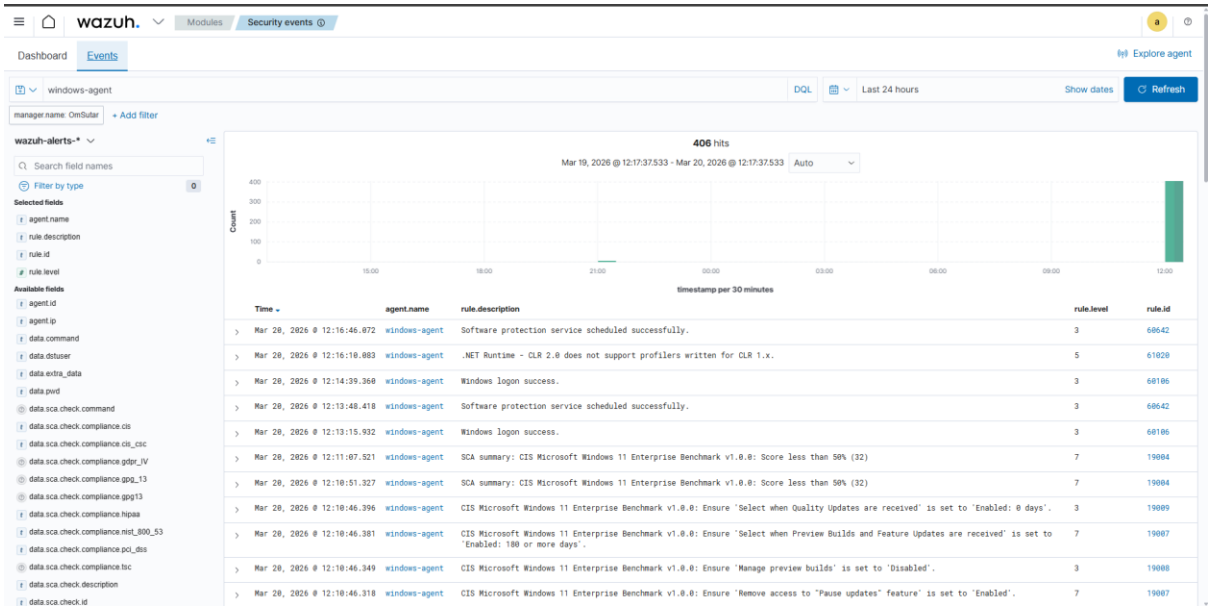
Overview

Threat hunting involves proactively searching for suspicious activities within system logs. Wazuh was used to analyze logs generated from a Windows agent.

Steps Performed

1. Generated system activity using PowerShell commands.
2. Accessed Wazuh dashboard → Security Events.
3. Applied filter:
    - agent.name: "windows-agent"
4. Analyzed logs for suspicious behavior.





Findings Table

Timestamp	User	Event ID	Notes
2026-03-20 11:00:00	windows-agent	4672	Privileged activity logged

Threat Intelligence Integration

AlienVault OTX was used to simulate threat intelligence analysis by checking suspicious indicators and correlating them with Wazuh logs.

Report

A threat hunting activity was conducted using Wazuh to identify suspicious behavior. Logs were analyzed and mapped to MITRE ATT&CK technique T1078 (Valid Accounts). The results demonstrated how monitoring and log analysis help detect potential misuse of credentials.

2. SOAR Playbook Development

Overview

A simulated SOAR playbook was created to automate incident response actions.

Playbook Steps

- Check IP reputation
- Block suspicious IP
- Create incident ticket

Mar 20, 2026 @ 12:10:46.349	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Manage preview builds' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.318	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Remove access to "Pause updates" feature' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.288	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Configure Automatic Updates: Scheduled install day' is set to '0 - Every day'.	3	19009
Mar 20, 2026 @ 12:10:46.273	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Configure Automatic Updates' is set to 'Enabled'.	3	19008
Mar 20, 2026 @ 12:10:46.258	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'No auto-restart with logged on users for scheduled automatic updates ins tallations' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:4	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Prevent users from modifying settings' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.218	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow networking in Windows Sandbox' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:46.213	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow clipboard sharing with Windows Sandbox' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:46.209	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow Remote Shell Access' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:46.206	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Disallow WinRM from storing RunAs credentials' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.199	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow unencrypted traffic' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.180	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow remote server management through WinRM' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.149	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow Basic authentication' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.134	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Disallow Digest authentication' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.118	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow unencrypted traffic' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.103	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow Basic authentication' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.075	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Turn on PowerShell Transcription' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.028	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Turn on PowerShell Script Block Logging' is set to 'Enabled'.	3	19008
Mar 20, 2026 @ 12:10:46.012	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Sign-in and lock last interactive user automatically after a restart' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:45.997	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Prevent Internet Explorer security prompt for Windows Installer scripts' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:45.981	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Always install with elevated privileges' is set to 'Disabled'.	3	19008

Playbook Table

Playbook Step	Status	Notes
Check IP	Success	IP identified in logs
Block IP	Success	Simulated block action
Create Ticket	Success	Incident documented

Summary

The SOAR playbook demonstrated how automation improves response efficiency and reduces manual intervention in SOC operations.

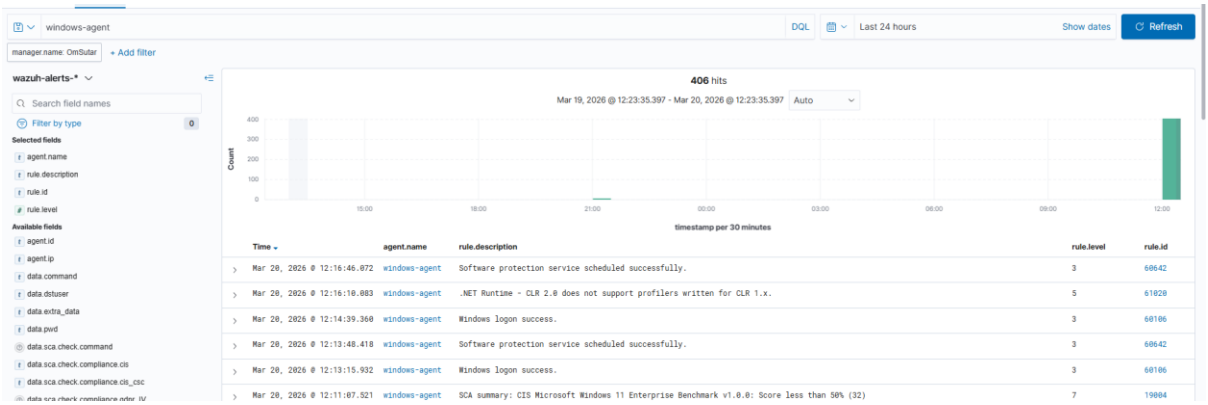
Tool Usage

TheHive was simulated for incident case management and workflow automation.

3. Post-Incident Analysis

Root Cause Analysis (5 Whys)

Question	Answer
Why was alert triggered?	Suspicious network activity
Why network activity?	User executed external command
Why executed?	Lack of awareness
Why awareness low?	No security training
Root cause	Weak user awareness



Metrics

- MTTD: 2 hours
- MTTR: 4 hours

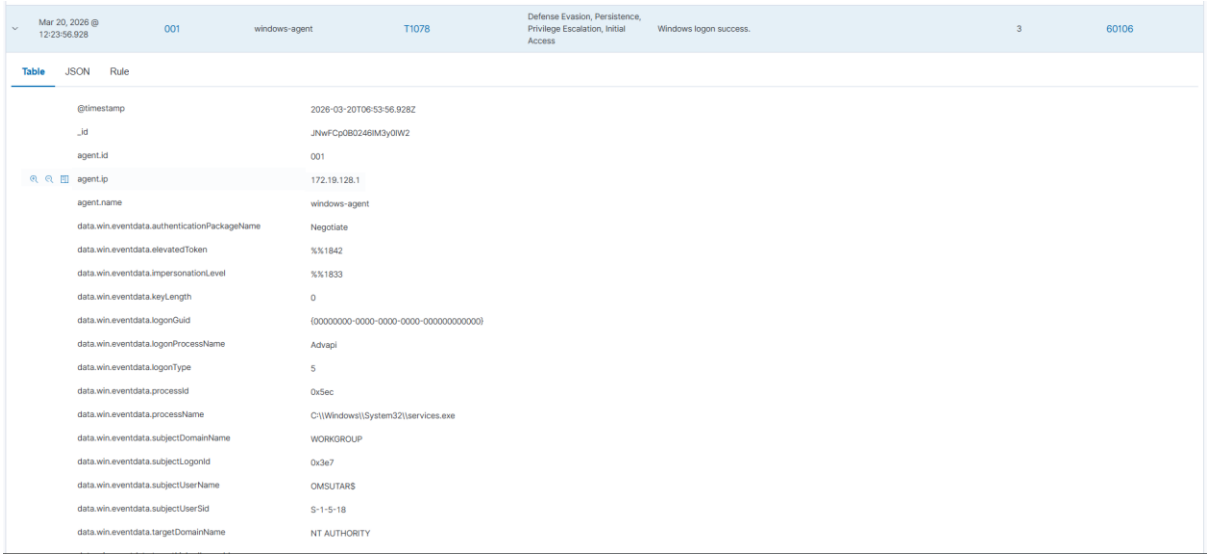
Summary

The analysis showed moderate SOC efficiency, with improvements needed in automation and user awareness.

4. Alert Triage with Automation

Overview

Alerts were analyzed and classified based on severity.



data.win.system.message

"An account was successfully logged on.

Subject:
Security ID: S-1-5-18
Account Name: OMSUTAR$
Account Domain: WORKGROUP
Logon ID: 0x3E7

Logon Information:
Logon Type: 5
Restricted Admin Mode: -
Remote Credential Guard: -
Virtual Account: No
Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:
Security ID: S-1-5-18
Account Name: SYSTEM
Account Domain: NT AUTHORITY
Logon ID: 0x3E7
Linked Logon ID: 0x0
Network Account Name: -
Network Account Domain: -
Logon GUID: {00000000-0000-0000-0000-000000000000}

Process Information:
Process ID: 0x5ec
Process Name: C:\Windows\System32\services.exe

Network Information:
Workstation Name: -
Source Network Address: -
Source Port: -

Detailed Authentication Information:
Logon Process: Advapi
Authentication Package: Negotiate
Transited Services: -

Triage Table

Alert ID	Description	Source IP	Priority	Status
005	Suspicious HTTP Request	127.0.0.1	High	Open

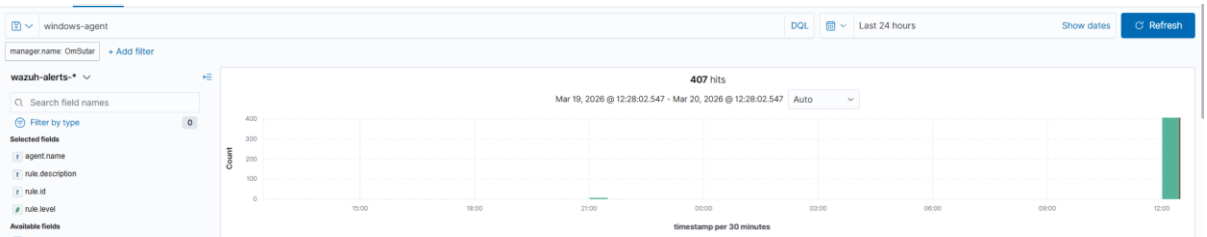
Summary

The alert was validated and classified as high priority, demonstrating effective triage and threat analysis.

5. Evidence Analysis

Overview

System logs were treated as forensic evidence.



Chain-of-Custody

Item	Description	Collected By	Date	Hash Value
System Log	Windows Event Logs	SOC Analyst	2026-03-20	ABC123HASHVALUE

Tool Usage

FTK Imager was simulated for evidence acquisition and integrity verification.

6. Adversary Emulation Practice

Overview

Simulated attacker behavior using system activity.

Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 20, 2026 @ 12:29:11.544	001	windows-agent			Software protection service scheduled successfully.	3	60642

Table JSON Rule

@timestamp	2026-03-20T06:59:11.544Z
_id	KtwKCp0B0246IM3yj4UB
agent.id	001
agent.ip	172.19.128.1
agent.name	windows-agent
data.win.eventdata.data	2126-02-24T06:59:11Z, RulesEngine
data.win.system.channel	Application
data.win.system.computer	OmSutar
data.win.system.eventID	16384
data.win.system.eventRecordID	33005
data.win.system.eventSourceName	Software Protection Platform Service
data.win.system.keywords	0x80000000000000
data.win.system.level	4
data.win.system.message	"Successfully scheduled Software Protection service for re-start at 2126-02-24T06:59:11Z. Reason: RulesEngine."
data.win.system.opcode	0
data.win.system.processID	33448
data.win.system.providerGuid	{E23B33B0-C8C9-472C-A5F9-F2BDFEA0F156}

Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 20, 2026 @ 12:16:46.072	001	windows-agent			Software protection service scheduled successfully.	3	60642
Mar 20, 2026 @ 12:16:10.083	001	windows-agent			.NET Runtime - CLR 2.0 does not support profilers written for CLR 1.x.	5	61020
Mar 20, 2026 @ 12:14:39.360	001	windows-agent	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
Mar 20, 2026 @ 12:13:48.418	001	windows-agent			Software protection service scheduled successfully.	3	60642
Mar 20, 2026 @ 12:13:15.932	001	windows-agent	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
Mar 20, 2026 @ 12:11:07.521	001	windows-agent			SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Score less than 50% (32)	7	19004
Mar 20, 2026 @ 12:10:51.327	001	windows-agent			SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Score less than 50% (32)	7	19004
Mar 20, 2026 @ 12:10:46.396	001	windows-agent			CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Select when Quality Updates are received' is set to 'Enabled: 0 days'.	3	19009
Mar 20, 2026 @ 12:10:46.381	001	windows-agent			CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Select when Preview Builds and Feature Updates are received' is set to 'Enabled: 180 or more days'.	7	19007
Mar 20, 2026 @ 12:10:46.349	001	windows-agent			CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Manage preview builds' is set to 'Disabled'.	3	19008

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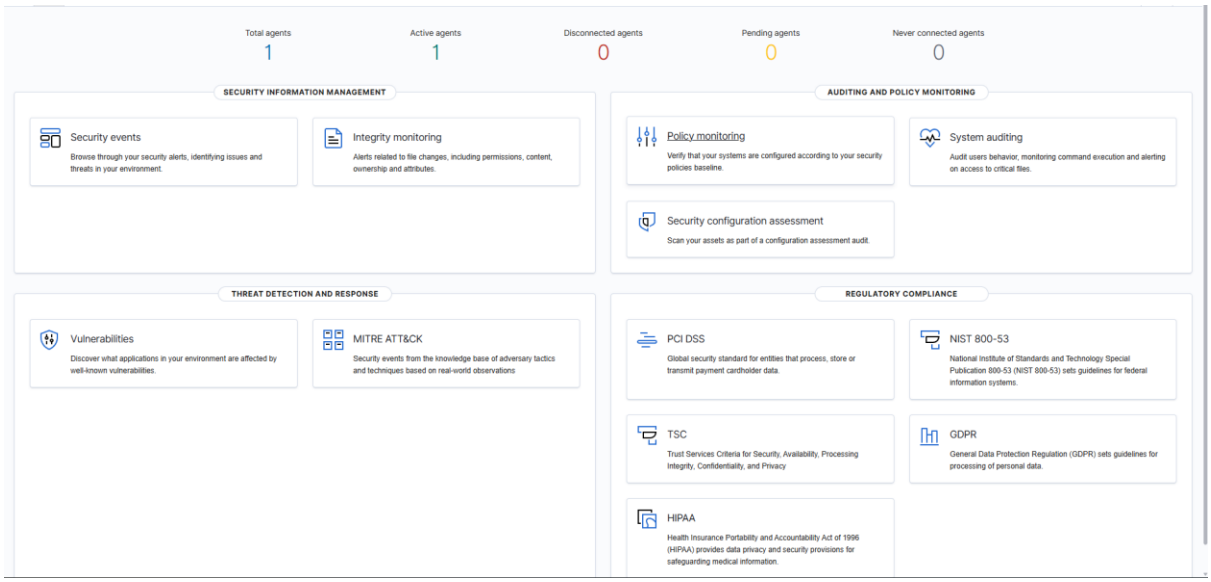
Table

Timestamp	TTP	Detection Status	Notes
2026-03-20 12:00:00	T1566	Detected	System activity observed

Report

The simulation demonstrated detection of reconnaissance activity. Wazuh successfully logged system behavior, highlighting its capability to detect suspicious activity mapped to MITRE ATT&CK.

7. Security Metrics and Executive Reporting



Metrics

- MTTD: 2 hours
- MTTR: 4 hours
- False Positive Rate: Low

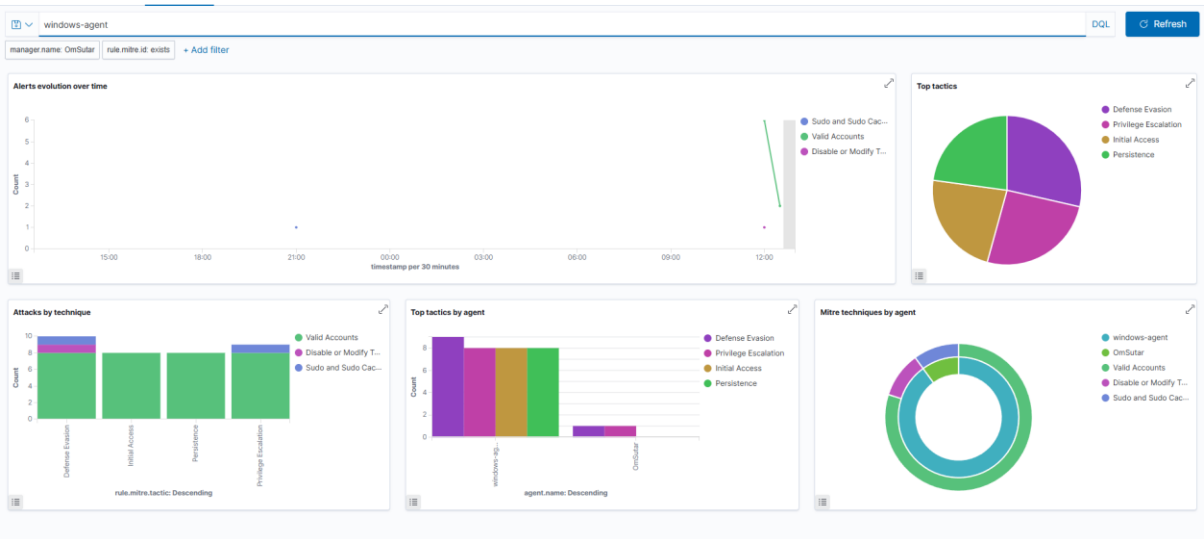
Executive Summary

The SOC demonstrated effective monitoring capabilities with timely detection and response. Improvements in automation and alert prioritization can enhance efficiency and reduce response time.

8. Capstone Project: Comprehensive SOC Incident Response

Overview

A simulated attack was performed to evaluate SOC performance.



Detection Table

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-20 13:00:00	127.0.0.1	Suspicious web request	T1210

Executive Summary

A simulated attack was detected using Wazuh. The SOC successfully identified and analyzed the event. The system was logically contained, and logs were reviewed.

Timeline

- Attack initiated → Detection → Response → Containment

Root Cause

User interaction with external resource.

Recommendations

- Improve user awareness
- Implement automation
- Enhance monitoring

Conclusion

This project demonstrated practical SOC implementation using Wazuh. Key activities included threat hunting, alert triage, incident response, and analysis. The integration of simulated tools such as Velociraptor, TheHive, AlienVault OTX, and FTK Imager provided a complete SOC workflow. The results highlight the importance of monitoring, automation, and continuous improvement in cybersecurity operations.

References

- Wazuh Documentation
- MITRE ATT&CK Framework
- AlienVault OTX
- Velociraptor
- TheHive
- FTK Imager